

Caffevolve — Phone-Primary Personalization

Addendum to “The Intelligence Center as a Distributed Process”: how resident profiles are reserved, finished, paired and kept in sync

Prepared for: Brent Langevin **Purpose:** Define which personalization tasks happen on the phone and which at the device, how a resident profile is reserved and paired, how edits stay in sync, and the commitments that keep the device useful on its own. **Status:** working draft, revised after review. The rules below were agreed in review; section 16 states the scope of this draft.

1. The core rule, in one line

The phone is the authoring surface; the device is the execution surface. Heavy personalization is done where the user is comfortable, and both surfaces may adjust what exists. All brew, grind and drink-path decisions, and every brew initiation, remain at the device.

2. Why

Personalization has grown: the initial profile, the Caffeine Behavior Instrument (CBI), subprofiles, BrewNow profiles, additive and grind preferences, and now daily plans and routines. Doing all of it at the device means long sessions standing at a countertop. Moving the heavy, optional authoring to the phone removes that friction and should reduce reluctance to use the personalization tools. The architecture already supports it: the engine logic is compiled into the app, and the per-user cloud store already carries the event log and the user’s parameters between the two copies of the engine.

3. Two stages of a resident profile

A Resident User Profile is built in two stages, by two different people.

Stage	Who	Where	What it holds
Basic Resident Profile	The administrator, behind the Admin PIN	At the device (Administrator Functions)	First name, initials, profile color and the user’s email address , which becomes the account’s user ID. It reserves one of the device’s profile slots.
Profile setup, producing the Complete Resident Profile	The user, with no administrator PIN	Device or phone	The CBI and the ceilings it derives, then the profile extras described in section 5.

- **Every resident profile starts this way, including the administrator's own.** The administrator first sets up the device (connection, beans, additives and the Admin PIN). No profile exists yet. The administrator then creates their own Basic Resident Profile, at that time or later, by the same route as everyone else.
- **The administrator never answers the CBI for anyone.** The slot is reserved by the administrator; everything personal is completed by the user.
- **Until the user finishes,** the profile appears as “Ready for you” with a “Finish setup” action, and the resident brewing paths (Home, Away, Plan my Day) stay locked. The user can brew through any guest channel in the meantime. The user may pause and resume setup, but cannot discard the reserved slot. Only the administrator removes a resident profile.
- **Resumable.** Setup already offers “Exit setup & complete later.” A user may answer part of the CBI on one surface and finish on either. The profile is established only when the CBI completes.
- **Bedtime is not part of the profile.** It is a session value, set and revisited at the device with each brew. It is never synced and cannot conflict.
- **Slots** are limited by tier (Originara 2, Advancara 5, Culminara Office 20), so the administrator decides who holds one.
- **Name, initials and color are per-device display settings.** They are not identifiers, are set by each device's administrator, and are not synced between a person's devices, so they may differ between home and office. Initials are 2–3 characters, which helps on a large Office device. A color already used by another profile on that device is unavailable, and the palette is unlimited, so every profile on a device stays distinct.

4. Reserving, inviting and the account

- **The email address is part of the administrator's step.** It is entered with the name, initials and color. The invite code is sent to it, and it becomes the user's account ID when the code is redeemed. It is not a credential. No phone number is collected anywhere.
- **The code.** On completing the Basic Resident Profile, the device registers the shell (name and email address) with the cloud and requests a one-time code, which is emailed to that address with a link to download the app. The device must be online at that moment. The user's personal data (CBI answers, ceilings, preferences) stays on the device until they pair.
- **The code is one-time, expires after a short window, locks after a few wrong attempts, and binds to the first account that redeems it.** The administrator can issue a fresh code, correcting the address if needed, if one expires or is lost.
- **Cleanup.** If a code expires unused, the shell's email address is deleted from the cloud. When the code is redeemed, the address becomes the account ID.
- **The account.** Redeeming the code and downloading the app creates the account. The invited email address is the user ID, and it is verified by the redemption itself. The user sets a

password. From then on, the email and password alone give access. The user then finishes the profile on the phone.

- **Recovery.** A forgotten password is reset by email, with no administrator involved. The password has minimum rules, and repeated failed sign-ins lock the account temporarily. A new or second phone is handled by signing in with the same email and password. Optional phone biometric unlock is a convenience after the first sign-in.
- **One account, one profile, bound permanently.** The claim cannot be undone or transferred, and there is no reassign function. The same profile works at home and at the office. If the wrong person claims a profile, for example because the address was mistyped, the administrator removes the profile and creates a new Basic Resident Profile. Removal deletes the cloud copy and unpairs the account.
- **Joining a second device.** The second device's administrator reserves a slot with the person's name, initials, color and the email address of their existing account, and issues the usual invite. The person redeems it while signed into that account, which attaches their existing profile to that device. It does not create a second profile, and no new CBI is needed. Each device an account uses consumes one slot on that device. The invited address must match the account's email.
- **An unpaired profile works fully at the device.** A local-only profile has no cloud backup, and a device reset or failure loses it. The app and the device say so when offering the invite.
- **Phone-first.** A person cannot self-create a resident profile on a household device. The administrator-issued invite is the only route to a slot, so one mechanism covers both directions.

5. What is created where

Item	Create	Edit
Basic Resident Profile: name, initials, color, email address (becomes the account ID)	Administrator, at the device (Admin PIN)	As today
Complete Resident Profile: CBI and the ceilings it derives	User, device or phone. Phone preferred. Resumable.	Either, through the profile review screen
Profile extras: additive preferences, grind setting, subprofiles, BrewNow profile, workplace settings	Phone only	Either
Today's plan (Plan my Day): build, or resume a routine for today	Either surface. Lasts for that day only and is not saved.	Either surface: add, change or remove any drink (time, type, amount), protect a drink, run Rebalance, or skip the plan
Routines (saved plans): create, edit, name, set saved days, delete	Phone only	Phone only

Item	Create	Edit
Guest profile (Quick Profile Path)	Device only	Device only

- **Complete on the device.** A resident can finish the CBI and ceilings at the device with no phone and no account. The device is never unusable for want of a phone. The device first-run screen says the longer setup is faster on the phone and offers to continue there.
- **Phone-only profile extras.** On the device, the profile setup screens for these extras say “available in the CaffeineGo app” and do not appear broken. Until the user sets them, the defaults are **no additives and a medium grind**.
- **Today’s plan versus routines.** A routine is a saved plan, kept for the days the user chooses (its **saved days**). Adjusting today’s plan never writes back to the routine. The device never saves or changes a routine, and runs Plan Check against whichever plan is in effect today.
- Each item is created in one place at a time, which removes most conflict risk. Remaining conflicts occur only on edits, covered in sections 7 and 8.

6. The boundary model

The Caffeine Comfort Zone has four parameters. Users can adjust only the two ceilings.

Parameter	Source	User-adjustable
Wake-Hours Ceiling	CBI, then refined by the AFS	Yes, any time, from the profile review screen (“Fine-tune ceilings”)
Bedtime Ceiling	CBI, then refined by the AFS	Yes, same as above
Peak-Zone Floor	CBI, then refined by the AFS	No
Caffeine Decay Rate	CBI, then refined by the AFS	No

- **Each ceiling has three parts.** The *computed value* (from the CBI, moved by the AFS); an optional *user override* (set with the slider); and the *effective value*, which is the override if one exists, otherwise the computed value. The device brews against the effective value. “Use computed” removes the override.
- **The platform only advises.** Setting a ceiling higher than most people are comfortable with shows an advisory message, not a block. The user decides.
- **The AFS never overwrites an override.** It continues to refine the computed value in the background, so “Use computed” returns the learned value. Its boundary-triggered prompts fire around the effective value.
- **Refinement is stored as feedback events.** Each feedback response is an immutable event with a source and ID, and the computed value is derived from the baseline plus those events. Events arrive from either surface and merge as a simple union, in any order, so the two surfaces cannot disagree about the computed value once they hold the same events and the

same engine version.

7. Editing and conflicts

The existing principle carries over: state changes are appended, not edited in place. Applied to profiles, subprofiles, preferences and plans:

- Every edit produces a **new version of the record**, stamped with its source (phone or device), time and the version it was based on.
- If both surfaces edited from the same base version, the **later edit wins per field**. A ceiling change on one side and a rename on the other both survive.
- If both changed the **same field**, the later edit wins and the other surface shows a one-line notice that its earlier change was replaced.
- The **two ceilings are treated as a group**, because they are adjusted together. If both surfaces edited them, the later edit wins for the pair, so the result is never a mix the user did not see.
- Each surface shows **sync state** (for example, “Synced to your Caffevolve 2 min ago”), especially after a ceiling change, because the device brews against whatever it last received.

For brew events the device remains the source of truth. For personalization records, the most recent valid version is authoritative, whichever surface wrote it.

8. Edit Profile: re-running setup

From the profile review screen, **Edit Profile** re-runs the whole setup flow, and everything in it can be changed: the CBI answers, subprofiles, grind setting and additive preferences. It is available on either surface. A re-run has two kinds of outcome:

- **CBI answers unchanged.** Only the other fields change, and they merge as ordinary field edits. The user’s ceiling overrides and the AFS learning are kept. Changing an additive must not wipe tuned ceilings.
- **CBI answers changed.** This is a new baseline. The computed boundaries and decay rate are recomputed, the user’s **ceiling overrides are cleared**, and the **AFS learning starts over**. The user passes through the fine-tune screen again and may set new overrides.
- Any edit made on the other surface against the old baseline is superseded by the new baseline, and that surface shows a notice.

9. Plan my Day and bedtime

Two different bedtimes are involved, and they are not the same thing.

- **Planned bedtime** is a required input when a plan or routine is set up. It lives with the plan, not the profile, and it is what the planner projects against.

- **Session bedtime** is set and confirmed at the device when a brew begins. It is not part of the profile. The session bedtime set at the device always rules over the planned bedtime.
- **The plan is re-projected first.** When a session bedtime is set, the device re-projects today's plan against it before anything else, whether the new bedtime is earlier or later. Drinks already consumed stay locked as the baseline; only the drinks not yet consumed are re-evaluated. The projection runs again every time a session bedtime is set.
- **Earlier bedtime:** the bedtime residual rises, and if it now exceeds the Bedtime Ceiling the planner offers Rebalance on the remaining drinks. **Later bedtime:** the residual falls, and the planner shows the new bedtime residual and the headroom.
- **The plan is never changed automatically.** The platform advises and the user decides. The session bedtime applies to today's plan only and never writes back to a saved routine's planned bedtime. The in-path reminder (FIG. 43D) includes the bedtime change in its planned-versus-current delta.

10. Guests

The phone-first rules apply to Resident Users only. The Guest path stays entirely on the device: name, initials, self-selected sensitivity level and bedtime, with no account, no invite and no CBI. A Returning Guest Profile continues to be kept for 72 hours.

11. Safety and access

- **The Admin PIN stays, and is unchanged in purpose.** It secures the device and decides who holds a profile on it. It covers device configuration (network, beans, canisters, hardware), reserving a slot, issuing an invite code, removing a resident profile, and resetting or transferring the device (which wipes local profiles and unpairs the device). One entry per visit to Administrator Functions.
- **The profile PIN** exists only for the shared Culminara Office device: a user-chosen four-digit PIN required when a profile is selected on an Office device. It only gates access to an Office device, and is never required on a home device, even though the same profile is used in both places. Profile changes made from the phone are made in the user's account with email and password, so it never applies there. The two PINs protect different things and never overlap.
- **Profile edits need no extra PIN.** Adjusting a ceiling, changing additives or re-running setup follows the device's rule: nothing at home, the profile PIN on an Office device.
- The platform continues only to advise. The user decides what to brew and what their ceilings are.

12. Design commitments

- **Minimal on-device path.** The device always supports the Complete Resident Profile (CBI and ceilings) and the Guest path, so it works without a phone.

- **Full local copy.** The device holds a complete copy of the active profiles and plans, so a cloud or app outage never strands a user's setup. Brewing continues from the last synced state.
- **The cloud remains a store, not a brain.** It holds the record and the current versions. It does not compute residual or decide what to brew.
- **Data governance defined before launch.** CBI answers describe sleep and caffeine habits and are health-adjacent, and the email address is personal data entered by one person about another. It is held in the cloud as the account ID once the invite is redeemed, and deleted if the invite expires unused. Consent, retention, export and account deletion rules need to be set, in addition to per-user isolation and encryption.
- **Version matching extended.** The requirement that firmware and app carry the same engine version also covers the record schema and the shared personalization flows, so older firmware does not meet newer app flows unprepared.
- **Validate the assumption.** Test completion rates for the CBI and planning on a phone prototype against the device before committing the full build.

13. Prototype status and remaining development

Done in prototype v2. An email address field (required, validated, unique per device) on the administrator's "Set Up a New User" step and on the first-profile step, hidden on Edit Profile. The administrator screen confirms that an invite code was emailed.

Deliberately left unchanged. The prototype still lets a whole resident profile, extras included, be set up at the device. This keeps the full profile available for demonstrations. Moving the extras to the app is a later change, listed below.

Remaining changes to complete the full process:

- **App-side account screens:** redeem the invite code, create the account (email as user ID, email verification, password), sign in, reset a password by email, and the profile setup and review screens in the app.
- **Join another device:** an existing account redeems an invite on a second device and attaches its profile, with no new CBI.
- **User profile setup flow at the device** (not device setup): the profile extras (additive preferences, grind setting, subprofiles, BrewNow profile) are created in the app. At the device these steps show "available in the CaffevolveGo app," with defaults of no additives and a medium grind. Editing existing extras stays available on either surface.
- **Plan my Day:** today's plan is built and adjusted on either surface (add, change, remove drinks; protect; Rebalance; skip) and is not saved; routines are created, edited, named, given saved days and deleted on the phone only. The planned bedtime is entered in the planning workspace, and the plan is re-projected whenever a session bedtime is set.
- **Figures:** FIG. 43A step 3802 (profile load excludes bedtime; planned bedtime entered in the workspace; re-projection step), "assign weekdays" (3832) and "matching the current

weekday” (4106) become saved days, and a phone authoring entry point is added.

- **Platform behavior to build:** issuing and redeeming invite codes, registering the profile shell with the cloud, account binding (one account, one profile), versioned per-field sync with ceilings treated as a group, AFS feedback stored as events, and the device’s full local copy.
- **Visitor pass (future concept):** the “Allow visitors” setting, pass issuing and expiry, the scan-or-tap flow, regular stops, and a list of places in the app (section 15, FIGS. 45A–B).
- **Platform Reference text:** the first profile is created by the administrator like any other; the invite and account model; one account holding one profile used at home and at work; the profile PIN applying to Office devices only; saved days; planned versus session bedtime; per-device name, initials and color.

14. Decisions recorded in review

- Name, initials and color are per-device display settings. Initials are 2–3 characters and a color is unique per device.
- The session bedtime set at the device always rules over a plan’s planned bedtime, and the plan is re-projected whenever it is set. Today’s instance only. A bedtime declared in CaffevolveGo counts the same way, because the app is an extension of the device; the most recent declaration applies.
- The Peak-Zone Floor and Caffeine Decay Rate are not user-adjustable. Defaults before the user sets them on the phone: no additives, medium grind.
- Today’s plan can be built and adjusted on either surface and is not saved. Routines are managed on the phone only.
- A profile is never moved to a different account. One account holds one profile, used at home and at the office.
- The user ID is the account email, entered by the administrator and verified when the user redeems the invite. A password reset goes by email, with no administrator. The platform holds no phone number.
- No profile PIN on home devices; the Admin PIN stays.
- The ceiling overrides and the AFS reset only when the CBI answers change.
- The AFS never overwrites a user’s override.

15. Future concept: the visitor pass

Working name: Caffevolve Away. The visitor pass would let a person use their one profile at any Caffevolve machine that has opted in, such as a café’s, with no profile slot, no invite and no account on the machine. It is beyond the current design. It is recorded here so the rest of the structure leaves room for it. FIG. 45A shows a first visit and FIG. 45B a regular stop.

- **The venue opts in.** The machine’s administrator turns on “Allow visitors” in Administrator Functions, behind the Admin PIN. Machines that have not opted in work as they do today.
- **The visitor grants a pass.** At the machine, the visitor scans or taps with the app. The app names the venue and what is shared, and the visitor allows it. The phone is the key, so no PIN is typed on a public machine.
- **A short-lived pass.** The cloud issues a pass bound to that one machine. It ends when the visitor walks away, after about two hours at most, or when they end it in the app.
- **Who knows what.** The machine shows a one-time code. The app answers it with the visitor’s signed-in account, and the cloud checks whether the visitor has a standing pass for that machine. With none, the app asks for consent and the cloud issues a one-visit pass. With one, a single tap confirms presence and the cloud issues the token. Either way the machine receives only a signed visit token saying what it covers and when it ends, so it behaves the same. The standing pass lives in the visitor’s account, not on the machine; removing a place in the app stops the cloud issuing tokens for it.
- **The machine borrows the profile.** It pulls the profile and today’s caffeine record, recomputes the standing with the same engine as every other device, and shows Plan Check and any open Sub-Profile. When the visit ends it deletes its copy.
- **What a visitor can do.** Brew, and adjust, protect or skip today’s plan. A visitor cannot change the profile. The machine never sees the email, the account or the CBI answers. The brew goes into the visitor’s own record, and the venue receives no personal data.
- **Regular stops.** For a place visited several times a week, the app offers to make it a regular stop. A standing pass is saved for that machine, so there is no daily pass and no consent screen. Each visit starts with one tap to confirm presence (where the venue charges, that tap is the approval of the total price), and the machine pulls everything fresh and keeps no profile data between visits. The standing pass ends when the user removes the place in the app, or lapses after about 90 days unused.
- **Without the app or a connection,** the Guest path remains.
- **Not the same as joining a second device.** That is membership: an administrator reserves a slot for a device the person uses regularly. The visitor pass is a guest arrangement for a machine the person does not belong to.
- **Managing passes in the app.** A “Places” list shows the regular stops, any visit in progress (with an “end this visit” button) and a short visit history. Adding, pausing or removing a standing pass is done there.
- **The machine adapts to itself.** A café machine has its own beans, additives and grind range. An additive in the visitor’s preferences that the machine does not carry is skipped, and the visitor is told. The caffeine blend is worked out for that machine’s beans, not the ones at home.
- **Bedtime is confirmed at the machine** for each brew, as everywhere. The app can pre-fill it from the visitor’s last declaration, because the app is an extension of the device.

- **A café brew is a measured entry.** The machine knows the exact dose, so it is more accurate than a drink logged by hand in the app, and the Adaptive Feedback System treats it as measured.
- **One visitor at a time per machine.** A visit ends when the visitor walks away or the pass expires, and the next visitor starts their own.
- **What it leaves unchanged:** slots, invites, one account holding one profile, the profile PIN (Office devices only), and the cloud as a store and not a brain.
- **Payment (planned for an upcoming version).¹**

¹ *Planned for an upcoming version: paying for the cup from the app, so that walking up, identifying yourself and brewing the recommended or planned cup is not undone by a queue at the till. The app always shows the total price and the visitor approves it before any payment starts; there is no automatic payment. On a first visit the approval is combined with allowing the pass, and at a regular stop it doubles as the presence tap. Payment goes through the phone's own wallet or an established payment provider, so Caffevolve stores no card details, and the money goes to the venue's own merchant account. At venues without a Caffevolve machine, paying from the app for what the user ordered, with no link to the venue's ordering system, is a possible later extension under the same approval rule. FIG. 46A shows the flow and FIG. 46B how it is set up in the app.*

16. Scope of this draft

This addendum fixes the structure and the general flow: who creates what, where, how a profile is reserved, paired and kept in sync, and how plans and bedtime behave across the phone and the device. Detailed policies and edge cases, such as password and lockout rules, device resale, invite delivery and optional planner advisories, are left for later versions of the specification. The current prototype does not yet show AFS learning state, sync or pairing, and still allows a whole profile to be set up at the device for demonstration; section 13 lists what remains. Section 15 records a future concept beyond the current design.

This addendum records the direction as agreed in review.